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Artificial intelligence at war

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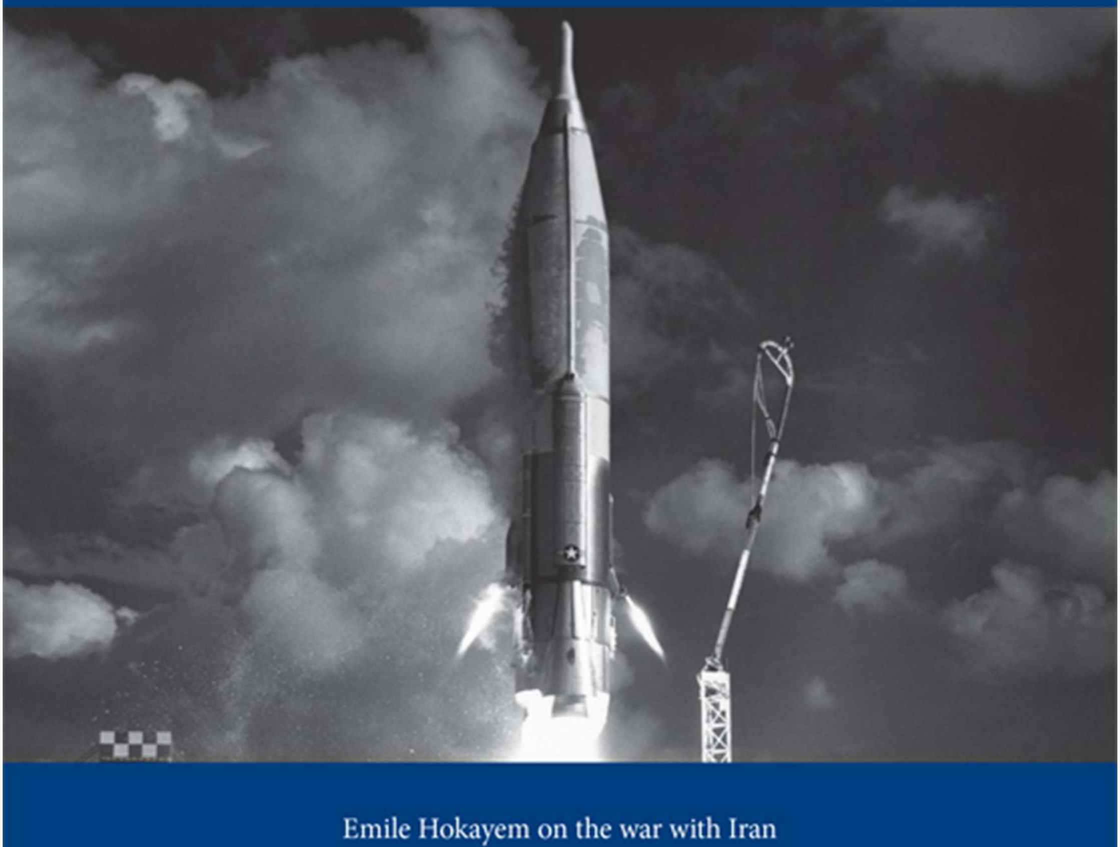

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Closing Argument

Strange Days: AI and the Next Industrial Revolution

Paul Fraioli

> The coal-output of the world, speaking roughly, doubled every ten years between 1840 and 1900, in the form of utilized power, for the ton of coal yielded three or four times as much power in 1900 as in 1840 … Since 1800 scores of new forces had been discovered; old forces had been raised to higher powers, as could be measured in the navy-gun; great regions of chemistry had been opened up, and connected with other regions of physics. Within ten years a new universe of force had been revealed in radiation. Complexity had extended itself on immense horizons, and arithmetical ratios were useless for any attempt at accuracy … But, at all events, the ten-year ratio seemed carefully conservative. Unless the calculator was prepared to be instantly overwhelmed by physical force and mental complexity, he must stop there.
>
> *Henry Adams,* The Education of Henry Adams *(1907)*[1]

I

The historian Henry Adams – grandson of one American president, great-grandson of another, educated in classical languages and statecraft, equipped with every advantage the nineteenth century could supply – was 62 in 1900 when he attended the Great Exposition in Paris. He went, he wrote, 'aching to absorb knowledge, and helpless to find it'. His friend

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Samuel Langley, the physicist and aviation pioneer, led him past the decorative exhibits and into the great hall of dynamos – the massive rotating machines that converted mechanical energy into electrical current, a technology then transforming cities and industry across the Western world. Adams was transfixed. 'He began to feel the forty-foot dynamos as a moral force, much as the early Christians felt the Cross.' He had spent a career studying the symbolic systems through which civilisations marshalled their energies – the Virgin Mary as the animating force of medieval Christendom, the crucifix as its rallying point – and here, humming at arm's length, was something somehow comparable. 'Before the end, one began to pray to it; inherited instinct taught the natural expression of man before silent and infinite force.'[2]

What troubled Adams was opacity. He knew intellectually that coal and steam in a far-off engine room yielded – via the disconcertingly quiet revolutions of the dynamo – an electrical current, but nonetheless felt the machine to be alien and 'anarchical'. The anxiety about technology that punctuates Adams's autobiography – expressed also with regard to X-rays, the Daimler motor and wireless telegraphy – had less to do with these machines themselves than his sense that they were mismatched to our intuitions.

Adams was witnessing the fruits of the Second Industrial Revolution. Its distinctive feature was that transformation proceeded not from a single invention, but from inventions in several domains combining and accelerating innovation across the board. Coal and steam had in an earlier period yielded railways and the factory. But when cheaper, mass-produced Bessemer steel became available alongside electrical power, the skyscraper and modern assembly line followed. Advances in metallurgy and the internal-combustion engine produced the automobile. Synthetic chemistry transformed agriculture and fuelled unprecedented population growth. Electrical power and telecommunications together created new forms of media, commerce and finance. Within roughly 40 years, daily life in industrialised countries changed dramatically. A more recent example is the smartphone: once mobile internet, global-positioning satellites, digital payments and cloud computing had matured, services such as ride-hailing applications emerged with surprising speed, obvious only in retrospect. This dynamic – with several adjunct technologies maturing and becoming cheap enough for the marketplace to embrace and combine – is again at work.

In December 2015, Klaus Schwab argued in *Foreign Affairs* that a 'fourth industrial revolution' was under way, driven by 'innovation based on combinations of technologies'.[3] Inspired in part by the work of economists Erik Brynjolfsson and Andrew McAfee, he was ahead of his time in important respects.[4] The argument rested on his observation that computing power was increasing exponentially, electronic data was proliferating and artificial intelligence (AI) was making 'impressive progress'. But a decade on, with far more advanced AI capabilities in hand, it is possible to see in greater detail the pathways to technological transformation in the coming years. Indeed, a revolution is now under way, propelled by advances in batteries, solar power, semiconductors, sensors and motors, with AI orchestrating new capabilities. Much of what follows could materialise by the late 2020s or early 2030s, a time frame well within the planning horizons of governments now in office.

Nowhere has this pattern been more visible, or unfolded more quickly, than with AI itself. In March 2026, Dario Amodei, the chief executive of Anthropic, predicted that this year would see a 'radical acceleration that surprises everyone'.[5] At the India AI Impact Summit in February 2026, Sam Altman of OpenAI said that 'we're going to have extremely capable models soon. It's going to be a faster takeoff than I originally thought.'[6] These claims invite scepticism, and many understand such pronouncements primarily as corporate hyperbole designed to inflate valuations and attract investment. The empirical record, however, does not support this interpretation.

The large language models powering AI have improved by a factor of 21 every two years since 2010, which long predates the race for investment capital, and the rate of improvement is not decelerating.[7] Public forecasts made by Altman, Amodei and other leaders from 2023 to 2025 have regularly been met or exceeded. Amodei in particular appears to regard his warnings about capability growth less as promotion than as candour directed at governments and publics he believes are dangerously complacent.

The central preoccupation in Silicon Valley since late 2025 has been the rapid maturation of AI agents: systems capable of sustained, multi-step autonomous action, qualitatively distinct from the chatbots of 2023.[8] Functioning increasingly like virtual remote workers, they can operate computers, use software tools and carry out complex work at superhuman speed.

As AI agents improve and global demand for them increases, so too will the need for processing power in the form of both electricity and inference compute. This intensifies a shift under way since 2024, before which AI models consumed most computing resources during the 'pre-training' phase of development. Inference compute, by contrast, is the computing power expended in response to individual queries. Despite massive public and private investment in American data centres and chip fabrication since 2023, even this may not be enough. More important still, frontier systems are becoming increasingly multimodal, able to interpret language, images, video, audio and other data streams, and to generate coordinated action across them. That is the precondition for moving from disembodied cognition to machines that can perceive, navigate and manipulate the physical world.

II

The consequences of advanced AI will be felt most acutely in the real world when it becomes embodied in tools, vehicles and robots. This path runs through a cluster of enabling technologies whose development conforms to a pattern familiar to manufacturing engineers: the S-curve. New products and processes usually advance slowly at first, as core technical problems are addressed and supply chains assembled. If they prove both technically and commercially viable, improvement accelerates: costs fall, performance rises and adoption spreads. In time, the curve begins to flatten as physical, economic or market limits are approached – at least until a fresh breakthrough begins another cycle.[9] The significance of the present moment is that several technologies central to embodied intelligence have, largely by coincidence, moved into or beyond the steep portions of their respective curves at roughly the same time.

Batteries are the clearest example. In 2010, lithium-ion battery packs cost roughly $1,500 per kilowatt-hour; today, they cost $108, while energy density has roughly tripled.[10] That shift made mass-market electric vehicles (EVs) viable. It also made possible a new class of untethered machines that can operate for meaningful periods without being connected to the grid. In Ukraine, the evolution of warfare in response to uncrewed capabilities has depended in part on precisely this kind of battery development. For robotics, the implication is simple enough: mobility and endurance are no longer the prohibitive constraints they once were.

Solar photovoltaics have undergone a comparable shift. In 1975, panels cost $128 per watt in real dollars, and in 2024, the price had fallen to $0.26 per watt.[11] Manufacturing has become cheap and relatively straightforward, above all in China, and in much of the world, solar is now the cheapest form of electricity generation ever deployed. That matters not only for decarbonisation, but also for computation and automation, both of which are ravenous for electricity. Data centres, EVs, robots and advanced manufacturing will demand vast quantities of power. Supplying it will be challenging in the short term, and innovation in solar-power generation and transmission will perforce accelerate.

Semiconductors entered the steep portion of their S-curve earlier as inputs to mass-market consumer electronics, but the rise of specialised computing architectures suited to neural networks has renewed their significance. NVIDIA was long known as a graphics-card company serving gamers. It turns out that chips designed to control tiny visual pixels require the same type of parallel computational power as neural-network training, so NVIDIA's processing architecture could be repurposed for machine learning. As a result, a single advanced graphics-processing cluster now delivers more computing power than the entire planet had in 2005. AI systems, in turn, have begun assisting in the design of better chips, creating the early stages of a recursive loop.

Sensors have followed a similar path, driven above all by economies of scale in the smartphone market. Light detection and ranging (LIDAR) systems, which use laser pulses to generate detailed three-dimensional maps of their surroundings, have fallen in cost from $75,000 in the early

2010s to under $200 today.[12] High-quality cameras that once cost thousands now cost a few dollars. These components were not originally developed for driverless vehicles, robots or advanced surgical systems, but are essential inputs that have only recently become accessible.

A less noticed but equally important advance lies in brushless electric motors and associated power electronics. In these systems, electronic controllers replace the mechanical contact brushes of older designs, improving efficiency and controllability. Motors relying on neodymium magnets – strong, compact magnets made from rare-earth elements and first developed in the 1980s – have increased in power density by a factor of at least ten in the last 20 years, with costs dropping precipitously.[13]

Taken separately, each development is important; together, they are foundational, enabling mass production of advanced and humanoid robots. Abundant, portable energy can provide these robots with power; semiconductors and AI can provide cognition and adaptability; sensors can provide perception; and motors, power electronics and actuators (which convert electricity into controlled movement) can provide the means to act. This matters because the most valuable robot is not necessarily the most specialised, but perhaps the one that can perform the most tasks without extensive redesign. A humanoid robot can work with tools, spaces and production systems already built for human beings. If such machines can be produced cheaply and in large numbers, they will become a general-purpose input for advanced manufacturing – operating equipment, moving components, assembling goods and, in time, helping to build more robots. Philip Trammell and Anton Korinek found in 2023 that fully automated robot production may ‘dramatically raise the growth rate and lower the labor share’ of the economy, overturning the long-standing pattern in which output, capital and the share of capital flowing to labour in advanced economies have tended to grow proportionally.[14]

In such a world, considerable advantages would accrue to the states or firms able to access power cheaply and manufacture components and finished robotics at scale, as part of an integrated industrial system. China, which installed roughly 54% of the world’s industrial robots in 2024 and whose firms have already brought low-cost humanoid platforms to market, enters this phase with a considerable advantage in industrial scale and as the global leader in open-weight AI models.[15] Its electricity generation is nearly triple that of the United States, a rough proxy for industrial capacity, and it refines roughly twice as much material as the rest of the world combined. Chinese firms already operate at a larger scale than those elsewhere in batteries, solar manufacturing, drone production and industrial robotics, and have moved quickly to lower the prices of sophisticated humanoid platforms.

In the West, no state comes close to matching China's hardware and manufacturing prowess, but one individual is attempting, through his companies, to gather comparable capabilities: Elon Musk. His companies span social media and communications (X Corp., formerly Twitter); frontier AI models and data centres (xAI); batteries, solar energy, EVs and robotics (Tesla); space launch (SpaceX); and broadband satellite internet (Starlink, a SpaceX subsidiary). Considered individually, these can look like the passion projects of an overextended entrepreneur. Together, they approach vertical integration across the core technologies likely to shape the next industrial era.

China enters this phase with a considerable advantage

In March 2025, Musk's AI company – which in two years he had built from nothing into a leading-edge developer of multimodal AI models and supercomputers – acquired X Corp. This converted the latter's data, users and live consumer base into resources for accelerating xAI's operations. The xAI facility in Memphis, Tennessee, known as Colossus, required 300 megawatts of power – as much as 250,000 households – to service 200,000 advanced graphics chips in March 2025.[16] Musk has described the sequence of improvisations required to secure that power: building a gas-turbine plant across the Mississippi state line when Tennessee permitting stalled, while confronting turbine-blade shortages severe enough that he expects SpaceX and Tesla may manufacture such components themselves.

On 2 February 2026, SpaceX acquired xAI. The rationale was again for one corporate entity to aid the other, with low-cost launch capabilities alleviating what Musk sees as the medium-term bottlenecks constraining the AI frontier: securing power for data centres and enough physical space on Earth to build them. His bet is that SpaceX's launch and satellite capabilities can be used to begin shifting AI infrastructure into orbit in 2028.[17] There, data centres would be powered by solar arrays, which are roughly five times more effective in space than on the ground, given the absence of both atmosphere and the day–night cycle, and avoid the permitting constraints and political conflicts faced by terrestrial data centres and energy systems. AI model weights and inference computation, in this view, would be transmitted back to Earth via satellite as physical computing infrastructure migrates outward. On a longer timescale, this idea extends to mining silicon and aluminium from the Moon.

Musk's largest firm, Tesla, has been pushing into new domains as well. In comments made this year, he described Tesla's success in developing self-driving EVs as a precursor technology – building out the manufacturing base, battery supply chains, motor expertise and machine-learning capabilities – for the mass production of humanoid robots. Musk has said there are 'three hard things' standing in the way of this: 'the real-world intelligence, the hand, and scale manufacturing'.[18] Optimus, Tesla's humanoid platform, merits attention for that reason. Its prototype hand has reportedly reached the full degrees of freedom of a nimble human hand, built with custom-designed actuators, motors, gears, sensors and power electronics. Notably, however, Optimus will not enter volume production before late 2026 at the earliest, and will remain in the flat stages of the S-curve for some time.

Musk's public claims evoke the obvious objection that he has a record of announcing timelines his companies then fail to meet, and that the distance between a prototype hand and a deployable industrial robot can be lengthened by thorny engineering problems and trade-offs. This objection is fair, but the precursor technologies have now matured to the point where the commercial logic of humanoid robotics is clear, and capital is beginning to flow accordingly. Musk may or may not be the one to achieve reliable production at scale. A Chinese rival or a less prominent firm may move faster. The more speculative ambitions, such as orbital data centres, may or may not prove feasible within the proposed timescales. What matters is that the underlying constraints have loosened significantly. The question to ask now is whether societies and institutions are prepared for what a successful programme of AI-driven industrial transformation, beginning in the late 2020s, would mean in the 2030s.

III

As AI solves some problems, others will arise. It holds genuine promise – in science, medicine, energy and the wider diffusion of prosperity – and it would be a mistake to treat what is coming as purely threatening. The economic risk most discussed today, particularly as AI agents mature, is labour-market disruption. But the larger and longer-term difficulties will be social and cultural, determining the habits, assumptions and tolerances of a given society well upstream of both politics and economics. Humans have long adapted to new tools, but have not had to accommodate the routine presence of machines that can perceive, converse, improvise and act in spaces previously reserved for people. The strangeness would lie not only in the displacement of particular forms of work, but in the gradual alteration of everyday life: the normalisation of artificial entities that appear responsive, useful and, at times, disconcertingly social; the fading of distinctions; and the arrival of a world in which competent agency, once scarce and therefore valuable, begins to seem abundant. In many societies, this would occur as populations decline due to collapsing birth rates.

Whether such a world will materialise is not assured. Advanced AI enlarges the possibility of more severe shocks, such as from biological terrorism or bio-engineering accidents. There are also the military dimensions, some of which are explored elsewhere in this issue. The confrontation in February between the US Department of War and Anthropic over the terms of the former's usage of the latter's AI systems – and the near-simultaneous deployment of Claude in the US–Israeli strikes on Iran – exposed in miniature a tension that will deepen. Frontier AI systems are becoming useful to war fighting more quickly than legal and ethical frameworks have been able to adapt, and governments (for both good and bad reasons) will likely show intense interest in the coming years in nationalising capabilities developed in the private sector. AI also makes mass surveillance trivially easy. Future humanoid robots, capable of operating in new environments with general-purpose intelligence, have obvious and significant military applications. But an acceptable humanoid age, as it were, is plausible only if graver perils are contained.

Returning to Adams, the nature of the man's weariness towards innovation was that, with each forward step, observers would be forced to undergo a re-education that would be, as he put it, 'violently coercive'. His self-image from the autobiography – of a man 'lying in the Gallery of Machines at the Great Exposition of 1900, his historical neck broken by the sudden irruption of forces totally new' – shows us as much. Yet Adams did not despair. He ended the book's most pointed chapter on this topic with an observation that reads today as both a salve and a warning: 'Thus far, since five or ten thousand years, the mind had successfully reacted [to technological acceleration], and nothing yet proved that it would fail to react – but it would need to jump.'[19]

Jump, or else what? Edmund Burke, writing as the French Revolution overturned a European order that had seemed settled, argued that institutions and social norms encode accumulated forms of practical wisdom that cannot safely be discarded in a moment of rationalist enthusiasm. One need not accept Burke's politics to recognise the force of the underlying point: the capacity of societies to absorb rapid change has limits, and beyond them lies convulsion rather than adaptation.

Social media was adopted by societies that could not anticipate its second-order effects. Within little more than a decade, the phenomenon had corroded public discourse, sharpened political polarisation, shortened attention spans and displaced forms of sustained association from which democratic societies draw resiliency. AI is different, but it is also a more powerful technology entering a world already weakened by prior experience. The risk is not simply that machine systems become more capable than the institutions around them, but that populations become accustomed to convenience, stimulation and algorithmic mediation, and surrender agency by increments too small to register until the habit of self-government has deteriorated.

The economist Tyler Cowen has observed that social changes often arrive by indirection rather than conquest. AI would be unlikely to displace human authority in some theatrical fashion.[20] The process would instead unfold through small acts of delegation: the government official relying on systems whose internal operations are too complex to interrogate with confidence; the manager or soldier accepting the machine's recommendation by default. Each individual step may appear modest. The cumulative effect may be to attenuate human judgement while preserving its outward forms. Societies will accommodate the strangeness of the artifice by running the Wizard of Oz's deception in reverse – placing a human face on machines and draping familiar social conventions over processes that are no longer human at all. It is unknown what will become of political accountability – or indeed of liberal democracy – when decisions are made by systems elected officials do not fully understand. You might argue, of course, that the arrival of such a phenomenon pre-dated AI advances this decade.

The cumulative effect may be to attenuate human judgement

Uncertainty about the international system's capacity to absorb what is coming tempts a kind of fatalism. Yet what has become clear in recent years is that the post-1945 order powerfully and effectively contained many of the explosive enmities now raging. Aspects of the liberal order served a dual role, both as an idea and a set of institutions that constrained the US from acting in its self-interest at all times, and as a polite fiction that enabled the US to do as it pleased when it felt the stakes were highest. The Trump administration has explicitly rejected this role. 'The days of the United States propping up the entire world order like Atlas are over', intones the November 2025 National Security Strategy.[21] It is an open question whether, in a period of sharp technological transition such as the one under way, a US operating within the liberal order could exploit its special position in the world – as the dominant source of frontier AI, the leading energy innovator, the seat of the firms building the next generation of robots – to achieve greater benefits for itself as well as its allies than one pursuing self-interest more narrowly defined.

The Second Industrial Revolution is generally considered to have ended in 1914. From an earlier vantage point, Adams could not have foreseen how cheap steel, electrical power, industrial chemistry and telecommunications would be leveraged for war, with an intense – but ultimately familiar – diplomatic tangle that summer, producing unprecedented slaughter. The future of AI is uncertain, but writing in early 2026, it strains credulity to imagine that the pallid pessimism prevailing in so much of the West – which longs for a bubble to burst or a wall to be hit – will prevail over the flush optimism in places like China and India. This does not mean that all the surprises ahead will be benign. The revolution now unfolding looks to be as significant as the Second Industrial Revolution, or more so. Adams's warning about acceleration thus remains in force: the mind will need to jump.

Notes

1 Henry Adams, *The Education of Henry Adams*, ed. Henry Cabot Lodge (Boston, MA: Massachusetts Historical Society, 1918), chapter 34.

2 Samuel Langley served as secretary of the Smithsonian Institution from 1887 to 1906. His friendship with Adams and their visit to the dynamo hall is recounted in *ibid.*, chapter 25.

3 Klaus Schwab, 'The Fourth Industrial Revolution: What It Means and How to Respond', *Foreign Affairs*, 12 December 2015, https://www.foreignaffairs.com/world/fourth-industrial-revolution.

4 See Erik Brynjolfsson and Andrew McAfee, *The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies* (New York: W. W. Norton, 2014).

5 Quoted in Supreeth Koundinya, 'Anthropic CEO Says 2026 Will Have a Radical Acceleration that Will Surprise Everyone', *Analytics India Magazine*, 4 March 2026, https://analyticsindiamag.com/ai-news/2026-will-have-a-radical-acceleration-that-surprises-everyone-anthropic-ceo.

6 Quoted in Divyanshi Sharma, 'Sam Altman at India AI Impact Summit 2026: 5 Key Highlights', Digit.in, 19 February 2026, https://www.digit.in/features/general/sam-altman-at-india-ai-impact-summit-2026-5-key-highlights.html.

7 Moore's Law is the observation that, since 1965, the transistor density of computer chips has improved at what is thought to be an extremely fast rate, with a doubling of transistor density every two years. This transformed the simple, typewriter-sized calculators of 1958 into a MacBook Air 50 years later. According to data tracked by Epoch AI, the computational resources used to train frontier AI models have

increased steadily since 2010 – with concomitant increases in capability – by a factor of 4.6 each year, or a factor of 21 every two years. See Epoch AI, 'AI Models', https://epoch.ai/data/ai-models.

8 See Paul Fraioli, 'India's AI Summit: A Success, but with Omissions', IISS Online Analysis, 26 February 2026, https://www.iiss.org/online-analysis/online-analysis/2026/02/indias-ai-summit-a-success-but-with-omissions/.

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10 Colin McKerracher, 'New Record Lows for Battery Prices', BloombergNEF, 19 December 2025, https://about.bnef.com/insights/clean-transport/new-record-lows-for-battery-prices/.

11 Our World in Data has compiled inflation-adjusted data since 1975 relying on three sources. See Our World in Data, 'Solar Photovoltaic Module Price', https://ourworldindata.org/grapher/solar-pv-prices.

12 See Matt McFarland, 'The $75,000 Problem for Self-driving Cars Is Going Away', *Washington Post*, 4 December 2015, https://www.washingtonpost.com/news/innovations/wp/2015/12/04/the-75000-problem-for-self-driving-cars-is-going-away/; and 'Exclusive: China's Hesai to Halve Lidar Prices Next Year, Sees Wide Adoption in Electric Cars', Reuters, 27 November 2024, https://www.reuters.com/technology/chinas-hesai-halve-lidar-prices-next-year-sees-wide-adoption-electric-cars-2024-11-27/.

13 See Akiya Kamimua, Sayako Sakama and Yutaka Tanaka, 'Characteristics of Hydraulic and Electric Servo Motors', *Actuators*, vol. 11, no. 1, January 2022, https://www.mdpi.com/2076-0825/11/1/11; and Alex Newkirk, Prakash Rao and Paul Sheaffer, 'U.S. Industrial and Commercial Motor System Market Assessment Report Volume 2: Advanced Motors and Drives Supply Chain Review', Lawrence Berkeley National Laboratory, September 2021, https://eta-publications.lbl.gov/sites/default/files/u.s._industrial_and_commercial_motor_system_market_assessment_report_volume_2-_advanced_motors_and_drives_supply_chain_review_a_newkirk_0.pdf.

14 Philip Trammell and Anton Korinek, 'Economic Growth Under Transformative AI', NBER Working Paper No. 31815, October 2023 (revised September 2025), https://www.nber.org/system/files/working_papers/w31815/w31815.pdf.

15 See International Federation of Robotics, 'World Robotics 2025 Report – INDUSTRIAL ROBOTS – Released by IFR', 25 September 2025, https://ifr.org/ifr-press-releases/news/global-robot-demand-in-factories-doubles-over-10-years; and Eric Schmidt and Selina Xu, 'China Could Dominate the Physical AI Future', *Time*, 3 March 2026, https://time.com/7382151/china-dominates-the-physical-ai-race/.

16 For context, the largest AI cluster in 2024, owned by Meta, contained 25,000 chips. Lennart Heim et al.,

'Trends in AI Supercomputers', arXiv:2504.16026, 23 April 2025, https://arxiv.org/abs/2504.16026.

17 Dwarkesh Patel, 'Elon Musk – "In 36 Months, the Cheapest Place to Put AI Will Be Space"', Dwarkesh Podcast, 5 February 2026, https://www.dwarkesh.com/p/elon-musk.

18 *Ibid*.

19 Adams, *The Education of Henry Adams*, chapters 25 and 34.

20 Tyler Cowen, 'The Surprising Truth Behind Collapsing Birthrates', Forecast 2050, 4 March 2026, https://podcastaddict.com/forecast-2050/episode/218533710.

21 White House, 'National Security Strategy of the United States of America', November 2025, p. 12, https://www.whitehouse.gov/wp-content/uploads/2025/12/2025-National-Security-Strategy.pdf.